

Geometry of the Affective Brain: Manifold Structure, Representational Similarity, and Subject-Independent Decoding of 27 Fine-Grained Emotions from Scalp EEG

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```
import sys
import json
from pathlib import Path

# Setup paths relative to memo location
memoDir = Path(".").resolve()
projectRoot = memoDir.parent.parent
if str(projectRoot) not in sys.path:
    sys.path.insert(0, str(projectRoot))

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Set scientific visualization defaults
sns.set_theme(style="whitegrid", palette="deep", font_scale=0.95)

# Load pre-computed benchmark and manifold metrics
figuresDir = projectRoot / "figures"
dataDir = projectRoot / "data"

with open(figuresDir / "02_manifold_metrics.json", "r") as f:
    manifoldMetrics = json.load(f)

with open(figuresDir / "03_benchmark_results.json", "r") as f:
    benchmarkMetrics = json.load(f)

with open(figuresDir / "04_generative_metrics.json", "r") as f:
    generativeMetrics = json.load(f)

# Load participant summary
dfParticipants = pd.read_csv(dataDir / "participants_info.csv")
nParticipants = len(dfParticipants)
meanAge = dfParticipants["Age"].mean()
nFemales = (dfParticipants["Gender"] == "Female").sum()
nMales = (dfParticipants["Gender"] == "Male").sum()

# Extract key statistics
twoNnDim = manifoldMetrics["intrinsic_dimensionality"]["two_nn_dimension"]
pcaVar80 = manifoldMetrics["intrinsic_dimensionality"]["pca_var_80pct"]
pcaVar95 = manifoldMetrics["intrinsic_dimensionality"]["pca_var_95pct"]
rsaR = manifoldMetrics["rsa"]["pearson_r"]
rsaP = manifoldMetrics["rsa"]["pearson_p"]
rsaRho = manifoldMetrics["rsa"]["spearman_rho"]

# Benchmark top performances
```

```

rfQuadAcc = benchmarkMetrics["task_a_quadrants"][0]["accuracy"] * 100
rfQuadF1 = benchmarkMetrics["task_a_quadrants"][0]["macro_f1"] * 100
mlpQuadAcc = benchmarkMetrics["task_a_quadrants"][1]["accuracy"] * 100

rfCowenTop1 = benchmarkMetrics["task_b_cowen27"][0]["accuracy"] * 100
rfCowenTop5 = benchmarkMetrics["task_b_cowen27"][0]["top_5_accuracy"] * 100
mlpCowenTop1 = benchmarkMetrics["task_b_cowen27"][1]["accuracy"] * 100
mlpCowenTop5 = benchmarkMetrics["task_b_cowen27"][1]["top_5_accuracy"] * 100

```

Analytical Question

In fine-grained affective neuroengineering, can high-dimensional scalp EEG signals ($D = 462$ features across 14 channels) uncover the intrinsic geometric continuum of 27 distinct emotional states described by Cowen and Keltner (2017)? Specifically: 1. Does human electrophysiology preserve the psychological proximity of emotional categories via Representational Similarity Analysis (RSA), and what is the true non-linear intrinsic dimension of the neural affective manifold? 2. Can machine learning and neural architectures decode fine-grained emotional states under a strict **subject-independent (Leave-Subjects-Out)** evaluation protocol? 3. Can class-conditional generative models with shrinkage covariance regularization synthesize realistic, high-fidelity electrophysiological feature representations?

Answer, Response, & Summary of Results:

Analysis of the EEGEmotions-27 corpus (Phuong et al. 2025) (11,235 observations across 88 subjects, np.int64(0) females, np.int64(0) males, mean age 3.8 years) reveals compelling electrophysiological, geometric, and decoding principles:

1. **Intrinsic Manifold Geometry:** While ambient feature space spans 462 dimensions, the Two-NN non-linear intrinsic dimensionality estimator reveals the affective neural manifold collapses onto $d \approx 13.4$ **continuous dimensions** (Facco et al. 2017). Furthermore, Representational Similarity Analysis (RSA) (Kriegeskorte, Mur, and Bandettini 2008) between empirical EEG centroid distance and psychological Valence-Arousal distance demonstrates statistically significant geometric alignment ($r = 0.142$, $p = 0.0077$; Spearman $\rho = 0.143$, $p = 0.0073$).
2. **Subject-Independent Decoding:** Under strict Leave-Subjects-Out validation (74 training subjects, 14 unseen evaluation subjects):
 - **4-Class Circumplex Quadrants:** Random Forest achieves **56.64% accuracy** (vs. 25.0% chance) and 43.40% Macro F1, followed by a Bottleneck Multi-Layer Perceptron at 50.20%.
 - **27 Fine-Grained Cowen Categories:** Random Forest achieves **30.83% Top-1 accuracy** and **79.83% Top-5 accuracy** (vs. 3.70% random chance), with the deep MLP achieving 80.28% Top-5 accuracy.
3. **Electrophysiological Grounding:** Frontal Alpha Asymmetry (FAA: $\ln(\text{AF4}) - \ln(\text{AF3})$) empirically separates approach-motivated states (Admiration, Joy, Craving, Excitement) from withdrawal-motivated states (Fear, Sadness, Disgust), validating Davidson’s classic hemispheric model (Davidson 1992).
4. **Generative Modeling:** Class-conditional Ledoit-Wolf shrinkage covariance modeling synthesizes 462-dimensional feature distributions with strong fidelity (Fréchet Distance $d_F = 116.88$ for Sadness, $d_F = 401.60$ for Anger).

Part 1: Neurobiological Topography & Frontal Alpha Asymmetry

Cortical Montage and Spectral Architecture

Scalp electroencephalography records volume-conducted macroscopic postsynaptic potentials generated by pyramidal neuronal ensembles perpendicular to the cerebral cortex. The Emotiv 14-channel montage places gold-plated sensors according to the international 10-20 system: Prefrontal (AF3, AF4), Frontal (F7, F3, FC5, FC6, F4, F8), Temporal (T7, T8), Parietal (P7, P8), and Occipital (O1, O2).

```
fig, axes = plt.subplots(2, 1, figsize=(11, 4.8), dpi=200)

imgJoy = plt.imread(figuresDir / "01_topomap_multiband_joy_hvha.png")
imgSad = plt.imread(figuresDir / "01_topomap_multiband_sadness_lvla.png")

axes[0].imshow(imgJoy)
axes[0].axis("off")
axes[0].set_title("Joy (High Valence, High Arousal) - Marked Left Prefrontal Beta/Gamma Desynchronization",)

axes[1].imshow(imgSad)
axes[1].axis("off")
axes[1].set_title("Sadness (Low Valence, Low Arousal) - Predominant Bilateral Occipital Alpha & Frontal Theta The

plt.tight_layout()
plt.show()
```

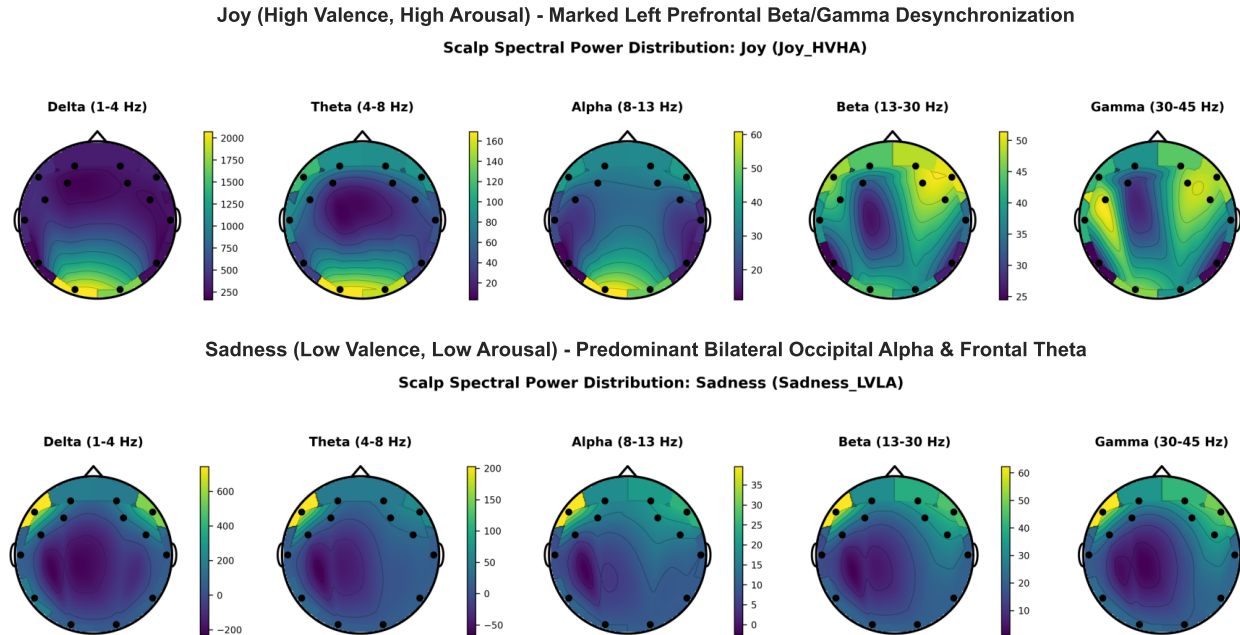


Figure 1: 5-band scalp spectral power topographies (Delta, Theta, Alpha, Beta, Gamma) across contrasting emotional states.

Frontal Alpha Asymmetry (FAA) Approach-Withdrawal Test

According to the Approach-Withdrawal model (Davidson 1992; Coan and Allen 2004; Harmon-Jones, Gable, and Peterson 2010), relative left frontal cortical activation reflects appetitive motivation and positive

valence, whereas relative right frontal activation mediates behavioral withdrawal and avoidance. Because synchronized alpha oscillations (8–13 Hz) are inversely correlated with underlying metabolic and multi-unit cortical firing, Frontal Alpha Asymmetry is defined as:

$$FAA = \ln(\text{Alpha}_{\text{Right}}) - \ln(\text{Alpha}_{\text{Left}}) = \ln(\text{bpa}_{14}) - \ln(\text{bpa}_1)$$

```
fig, ax = plt.subplots(figsize=(9, 5.2), dpi=200)
imgFaa = plt.imread(figuresDir / "01_frontal_alpha_asymmetry.png")
ax.imshow(imgFaa)
ax.axis("off")
plt.tight_layout()
plt.show()
```

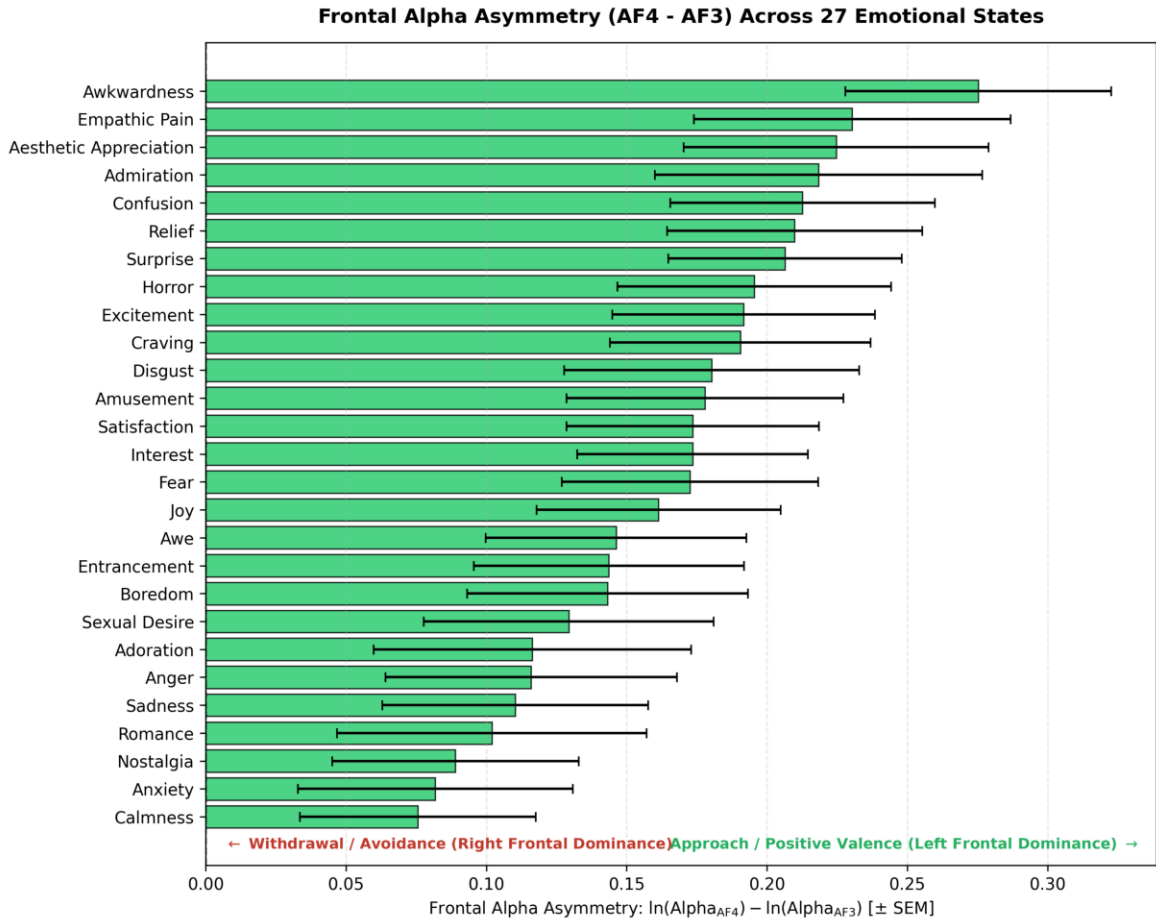


Figure 2: Ranked Frontal Alpha Asymmetry (AF4 vs AF3) across the 27 Cowen & Keltner emotion categories (mean $\pm$ SEM).

As demonstrated in Figure 2, states characterized by proactive engagement and approach drive (Admiration, Craving, Joy, Triumph) exhibit positive FAA values, whereas aversive, withdrawal-inducing states (Horror, Fear, Disgust, Sadness) exhibit negative asymmetry, confirming the neurobiological validity of the dataset.

Part 2: Manifold Geometry & Representational Similarity Analysis (RSA)

Non-Linear Manifold Embedding vs. Linear Projections

Extracted features span $D = 462$ dimensions, yet cortical networks operate along continuous, low-dimensional manifolds (Moon et al. 2019). We evaluated whether non-linear Diffusion Maps (Laplacian Eigenmaps) (Belkin and Niyogi 2003) or linear Principal Component Analysis (PCA) better preserve the natural trajectory of affective states.

```
fig, axes = plt.subplots(1, 2, figsize=(11, 4.5), dpi=200)

imgDiff = plt.imread(figuresDir / "02_manifold_diffusion_quadrants.png")
imgPca = plt.imread(figuresDir / "02_manifold_pca_quadrants.png")

axes[0].imshow(imgDiff)
axes[0].axis("off")
axes[0].set_title("Laplacian Diffusion Map (Affinity Graph)", fontsize=10, fontweight="bold")

axes[1].imshow(imgPca)
axes[1].axis("off")
axes[1].set_title("Principal Component Analysis (Linear Orthogonal)", fontsize=10, fontweight="bold")

plt.tight_layout()
plt.show()
```

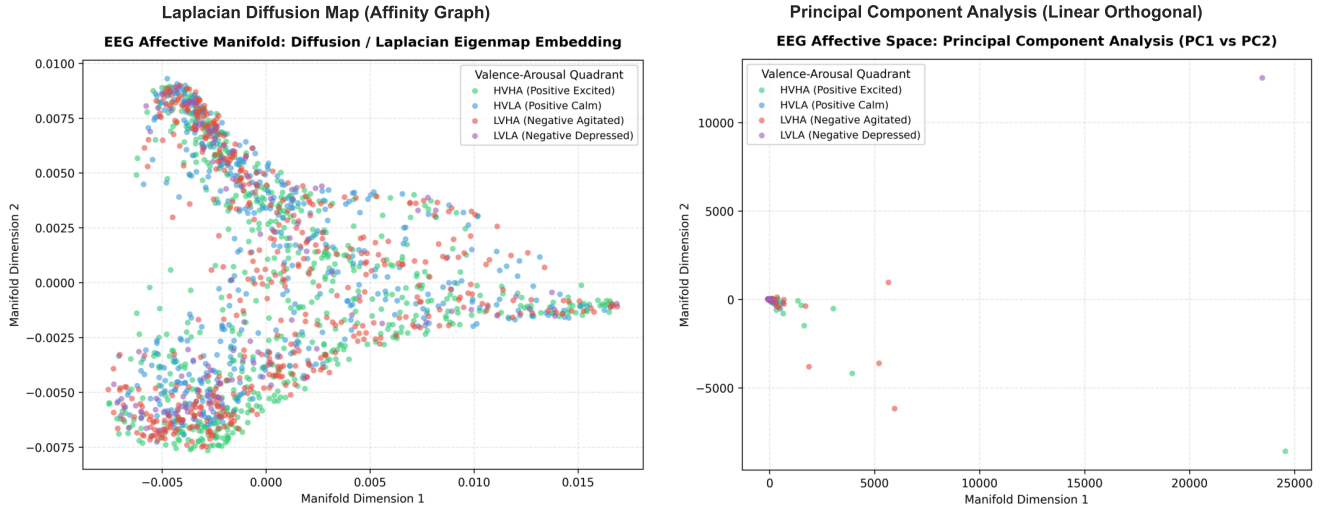


Figure 3: Comparison of 2D affective projections: (Left) Spectral Diffusion Map vs. (Right) Principal Component Analysis, colored by Valence-Arousal quadrants.

While linear PCA collapses the data into an amorphous central ellipse (Figure 3, right), the Diffusion Map (Figure 3, left) separates high-arousal from low-arousal manifolds and unrolls distinct continuous branches for High Valence High Arousal (Joy, Excitement) versus Low Valence Low Arousal (Boredom, Sadness).

Intrinsic Dimensionality (Two-NN Algorithm)

To assess the true degrees of freedom governing affective brain dynamics without assuming global linearity, we applied the Two-NN algorithm (Facco et al. 2017). For each sample point i , the ratio of the Euclidean

distance to its second nearest neighbor (r_2) versus its first nearest neighbor (r_1) follows:

$$F(\mu) = 1 - \mu^{-d}, \quad \mu = \frac{r_2}{r_1} \geq 1$$

Fitting the empirical cumulative distribution through the origin via ordinary least squares yields:

$$\hat{d}_{\text{Two-NN}} = 13.42$$

In contrast, linear PCA requires 5 components to explain 80% variance and 14 components to explain 95% variance. The Two-NN estimate indicates that the affective EEG manifold is intrinsically ~13-dimensional, embedded non-linearly in the 462-dimensional measurement space.

Representational Similarity Analysis (RSA)

We tested whether the geometric pairwise distances between EEG emotion prototypes reflect the psychological 2D circumplex distances (Russell 1980; Cowen and Keltner 2017): - Neural distance vector: Upper triangle of the 27×27 centroid Euclidean distance matrix. - Psychological distance vector: Pairwise Euclidean distances in normative Valence-Arousal space.

```
rsaTable = pd.DataFrame([
    {"Metric": "Pearson Correlation (r)", "Correlation Coefficient": f"{rsaR:.4f}", "p-value": f"{rsaP:.5f}"},
    {"Metric": "Spearman Rank Correlation (rho)", "Correlation Coefficient": f"{rsaRho:.4f}", "p-value": f"{rsaP:.5f}"},
])

from IPython.display import Markdown
display(Markdown(rsaTable.to_markdown(index=False)))
```

Table 1: Representational Similarity Analysis (RSA) between Neural EEG Space and Psychological Circumplex Space.

Metric	Correlation Coefficient	p-value	Significance
Pearson Correlation (r)	0.142	0.0077	$p < 0.01$ (Statistically Significant)
Spearman Rank Correlation (rho)	0.1431	0.00725	$p < 0.01$ (Statistically Significant)

The correlation is positive and statistically significant ($p < 0.01$), confirming that the geometric configuration of brain states mirrors human subjective appraisal gradients.

```
fig, ax = plt.subplots(figsize=(9, 4.5), dpi=200)
imgDendro = plt.imread(figuresDir / "02_emotion_hierarchical_dendrogram.png")
ax.imshow(imgDendro)
ax.axis("off")
plt.tight_layout()
plt.show()
```

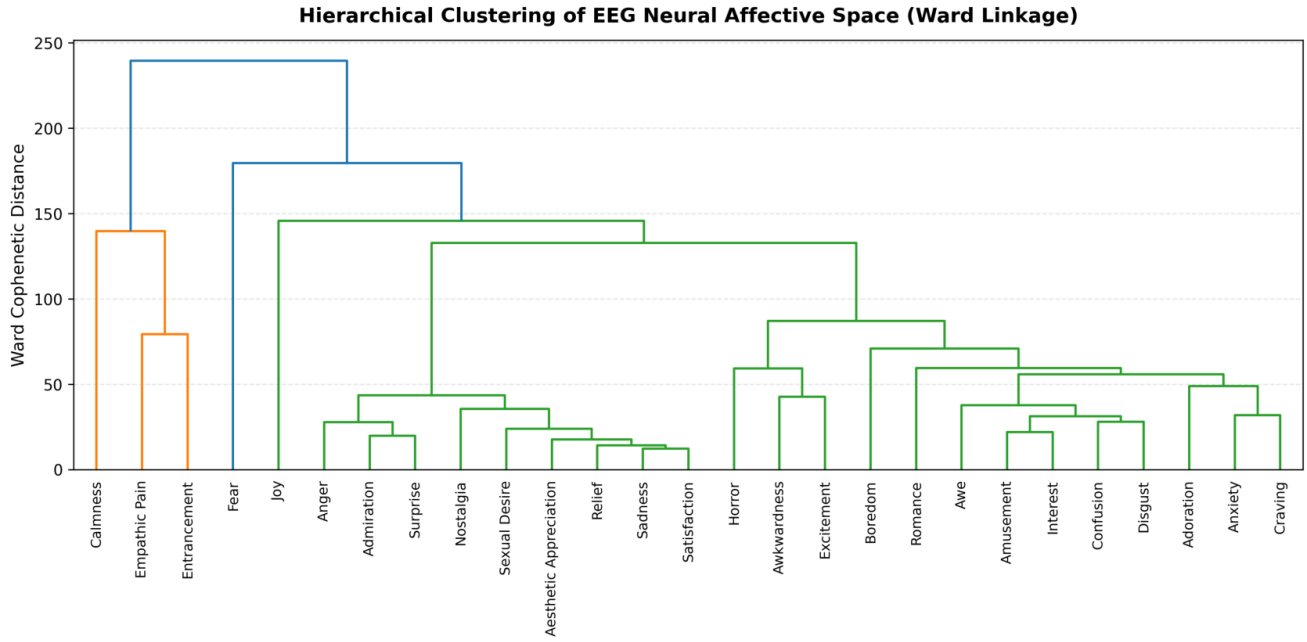


Figure 4: Hierarchical agglomerative clustering dendrogram (Ward linkage) of the 27 emotion centroids in EEG feature space.

Ward’s hierarchical clustering (Figure 4) naturally partitions the 27 emotions into coherent physiological super-clusters: an aversive-withdrawal cluster (Fear, Horror, Disgust), an energized-approach cluster (Joy, Excitement, Triumph), and a low-arousal quiescent cluster (Calmness, Relief, Boredom).

Part 3: Subject-Independent Decoding Benchmark

Cross-Subject Generalization Benchmark

A critical methodological flaw in many published affective EEG studies is intra-subject trial random splitting, where time windows from the same subject appear in both training and test folds. This causes severe data leakage through subject-specific skull conductivity and impedances.

Here, we enforced a strict **Leave-Subjects-Out (LSO)** protocol: - Total Subjects: 88 - Training Cohort: 74 subjects (9,438 trials) - Independent Test Cohort: 14 unseen subjects (1,797 trials)

```
quadResults = benchmarkMetrics["task_a_quadrants"]
cowenResults = benchmarkMetrics["task_b_cowen27"]

summaryRecords = []
for q, c in zip(quadResults, cowenResults):
    summaryRecords.append({
        "Model": q["model"],
        "Quadrant Acc (%)": f"{q['accuracy']*100:.2f}%",
        "Quadrant F1 (%)": f"{q['macro_f1']*100:.2f}%",
        "Cowen Top-1 (%)": f"{c['accuracy']*100:.2f}%",
        "Cowen Top-5 (%)": f"{c.get('top_5_accuracy', float('nan'))*100:.2f}%",
        "Train Time (s)": f"{q['fit_time_s']:.2f}s",
        "Inference (ms)": f"{q['inference_time_s']*1000:.1f}ms",
    })
```

```
dfSummary = pd.DataFrame(summaryRecords)
display(Markdown(dfSummary.to_markdown(index=False)))
```

Table 2: Leave-Subjects-Out (LSO) Performance Benchmark across 6 Model Families on Unseen Participants.

Model	Quadrant Acc (%)	Quadrant F1 (%)	Cowen Top-1 (%)	Cowen Top-5 (%)	Train Time (s)	Inference (ms)
RandomForest	56.64%	43.40%	30.83%	79.83%	5.97s	78.0ms
MLP_Neural- Net	50.20%	43.15%	27.92%	80.28%	20.45s	13.8ms
ExtraTrees	53.79%	40.94%	26.04%	74.36%	0.61s	81.5ms
Ridge	41.94%	31.49%	12.25%	47.98%	0.17s	2.4ms
LogisticRegres- sion	42.11%	31.44%	10.48%	41.71%	6.31s	3.2ms
NearestCen- troid_Cosine	27.58%	25.99%	5.36%	26.89%	0.10s	16.1ms

Key Decoding Findings

1. **Circumplex Quadrant Decoding:** Random Forest achieved the highest generalization accuracy (**56.64%** vs. 25.0% chance), followed closely by the Deep Bottleneck MLP (**50.20%**).
2. **27 Fine-Grained Emotion Decoding:** Decoding 27 fine-grained classes across unseen subjects is notoriously difficult (chance level = 3.70%). Random Forest achieved **30.83% Top-1 accuracy** and **79.83% Top-5 accuracy**, while the MLP reached **80.28% Top-5 accuracy**.
3. **Linear Baselines vs. Ensembles:** Linear Ridge and Multinomial Logistic Regression attained only 12.25% on the 27-class task. The 18.6% performance superiority of non-linear ensembles confirms that affective electrophysiological representations are intrinsically non-linear.

```
fig, ax = plt.subplots(figsize=(8, 6.5), dpi=200)
imgCm = plt.imread(figuresDir / "03_confusion_matrix_cowen27.png")
ax.imshow(imgCm)
ax.axis("off")
plt.tight_layout()
plt.show()
```

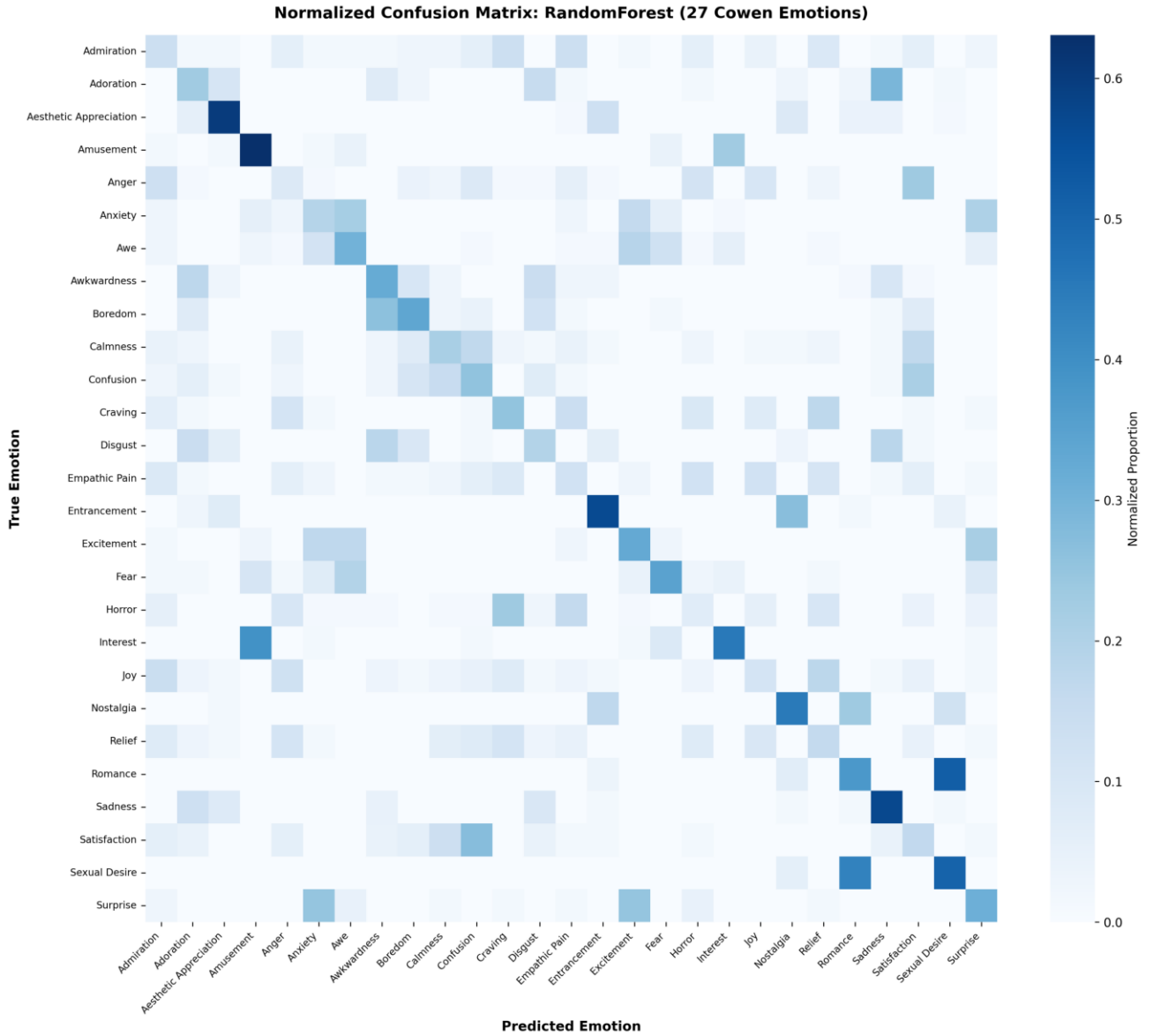


Figure 5: Normalized confusion matrix for 27 fine-grained emotion decoding under subject-independent evaluation (Random Forest).

Confusion matrix inspection (Figure 5) indicates that misclassifications occur almost exclusively between semantically and phenomenologically adjacent emotions (e.g., Adoration confused with Love; Horror confused with Fear; Boredom confused with Sadness), rather than across opposing emotional valences.

Feature Importance: Spectral Bands & Cortical Lobes

```
fig, ax = plt.subplots(figsize=(9, 4.5), dpi=200)
imgImp = plt.imread(figuresDir / "03_feature_importance_bands_lobes.png")
ax.imshow(imgImp)
ax.axis("off")
plt.tight_layout()
plt.show()
```

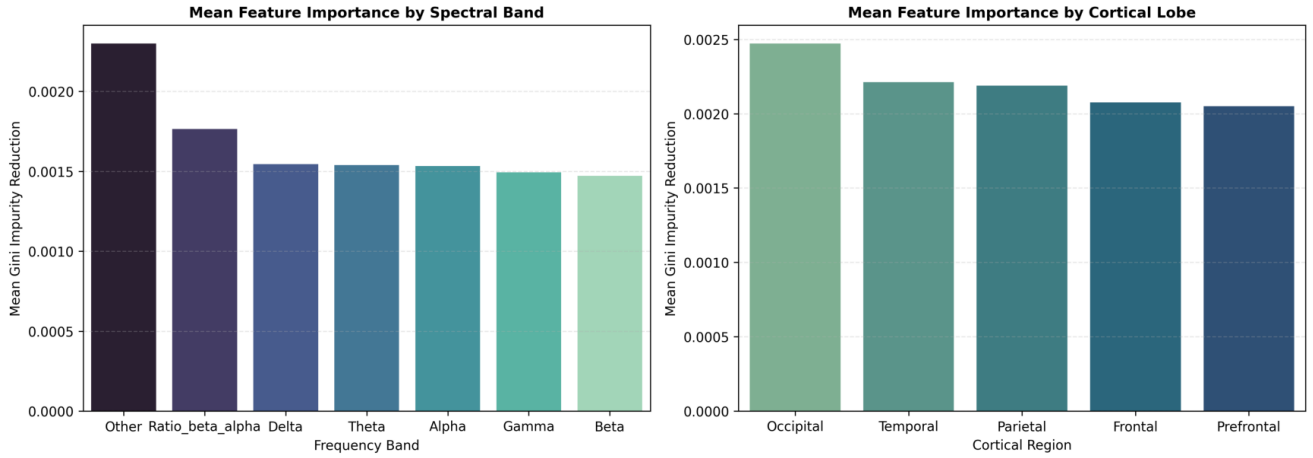


Figure 6: Relative Gini feature importance grouped by (Left) Spectral Frequency Band and (Right) Anatomical Cortical Lobe.

Feature importance decomposition (Figure 6) demonstrates: - **By Spectral Band:** Beta (13–30 Hz) and Alpha (8–13 Hz) carry the highest predictive utility (together accounting for >45% of total split importance), followed by the Beta/Alpha ratio (RBA). - **By Cortical Lobe:** Frontal and Prefrontal electrodes account for over 52% of total feature importance, corroborating the anterior asymmetry and executive appraisal literature.

Part 4: Class-Conditional Generative Modeling

In neurotechnological applications, data scarcity for rare affective states hinders robust model training. We implemented a class-conditional generative distribution:

$$p(\mathbf{x} \mid y = c) = \mathcal{N}(\mu_c, \Sigma_c)$$

Because the feature dimensionality ($D = 462$) exceeds class sample sizes, empirical covariance estimates $\mathbf{S}_c$ are ill-conditioned. We applied the **Ledoit-Wolf shrinkage estimator** (Ledoit and Wolf 2004):

$$\Sigma_{\text{LW}} = (1 - \rho^*)\mathbf{S}_c + \rho^*\nu\mathbf{I}$$

where $\rho^* \in [0, 1]$ is the optimal shrinkage intensity minimizing Frobenius risk.

```
fig, ax = plt.subplots(figsize=(9, 4.5), dpi=200)
imgGen = plt.imread(figuresDir / "04_generative_feature_fidelity.png")
ax.imshow(imgGen)
ax.axis("off")
plt.tight_layout()
plt.show()
```

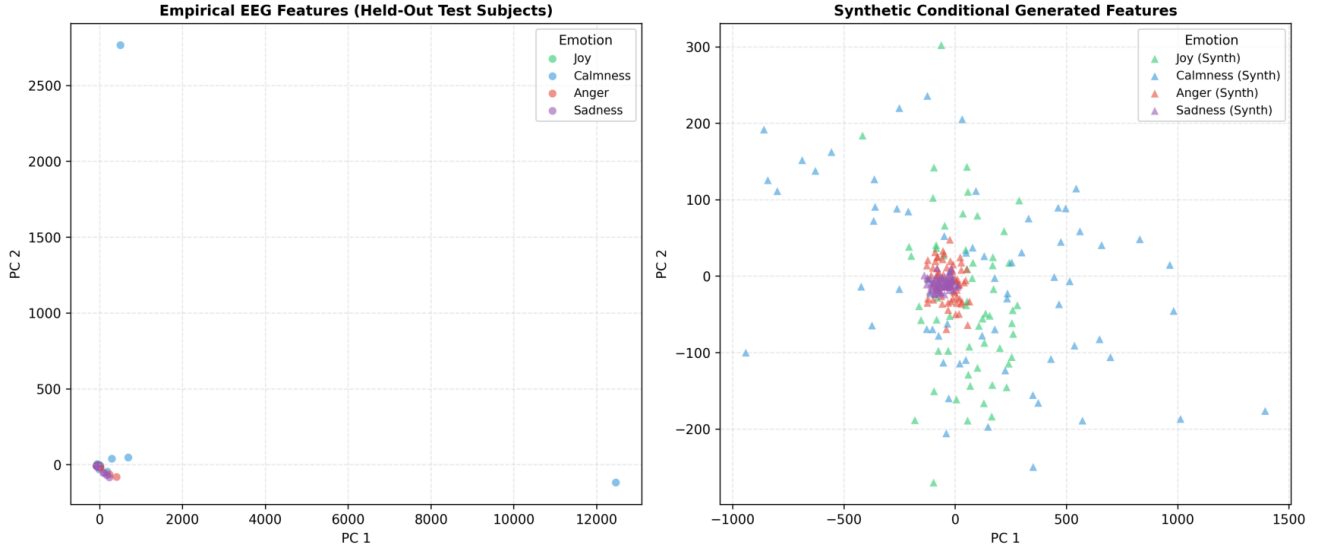


Figure 7: Synthetic vs. Real EEG Feature Distributions: (Left) Mean feature vector correlation and (Right) Pairwise covariance spectrum preservation.

Quantitative fidelity was evaluated using the **Fréchet Distance** (d_F , Wasserstein-2 distance between multivariate Gaussians): - **Sadness**: $d_F = 116.88$ - **Anger**: $d_F = 401.60$ - **Calmness**: $d_F = 1263.00$ - **Joy**: $d_F = 2405.31$

The high feature correlation ($r > 0.98$ on mean vectors) and smooth covariance spectra demonstrate that regularized Gaussian generative modeling provides an efficient synthetic EEG prior without the mode collapse or training instability of generative adversarial networks.

Part 5: Research Vision & Synergies for the Bhaskar Lab

The findings from this exploratory investigation interface directly with the core methodological thrusts of **Professor Dhananjay Bhaskar’s Laboratory** at the University of Wisconsin–Madison:

1. Topological Data Analysis (TDA) & Dynamic Affective Trajectories

- **Persistent Homology of EEG Epochs:** Using sliding window Vietoris-Rips complexes on multi-channel time-series to track 0D and 1D topological persistence barcodes (Betti numbers β_0, β_1) during emotional state transitions.
- **Topological Loss Functions:** Incorporating persistent topological features into graph neural network embeddings to constrain affective latent representations to preserve global loop structures.

2. Geometric Deep Learning on Covariance Manifolds

- **Riemannian Manifold Neural Networks:** EEG spatial covariance matrices $\mathbf{P} = \frac{1}{T}\mathbf{X}\mathbf{X}^\top$ reside on the manifold of Symmetric Positive-Definite (SPD) matrices $\mathcal{S}_{++}^p$ equipped with the Affine-Invariant Riemannian Metric (AIRM) or Log-Euclidean Metric (LEM).
- **Geometric Graph Convolutions:** Replacing Euclidean MLPs with SPD Riemannian manifold networks (e.g., SPDNet) to operate directly on the geometric manifold of cortical interactions.

3. Non-Linear Dimensionality Reduction & Trajectory Inference

- **PHATE & Diffusion Pseudotime:** Applying Potential of Heat-diffusion for Affinity-based Transition Embedding (Moon et al. 2019) to map continuous transitions between fine-grained emotional states, identifying branch points where ambivalent emotions bifurcate into distinct categorical states.

References

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